ReXVQA: A Large-scale Visual Question Answering Benchmark for Generalist Chest X-ray Understanding

Paper

Dataset

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Motivation

1. Current CXR AI approaches operate within narrow constraints, limited to predetermined pathology classification rather than engaging in the broader analytical reasoning that characterizes expert radiological assessment.
2. Existing medical VQA datasets rely heavily on templated question generation and lack the diversity and complexity of real clinical reasoning, limiting their effectiveness as generalist benchmarks.
3. Systematic evaluation of multimodal LLMs across clinically meaningful tasks remains challenging, preventing assessment of their capabilities for flexible visual reasoning and natural dialogue in clinical practice.

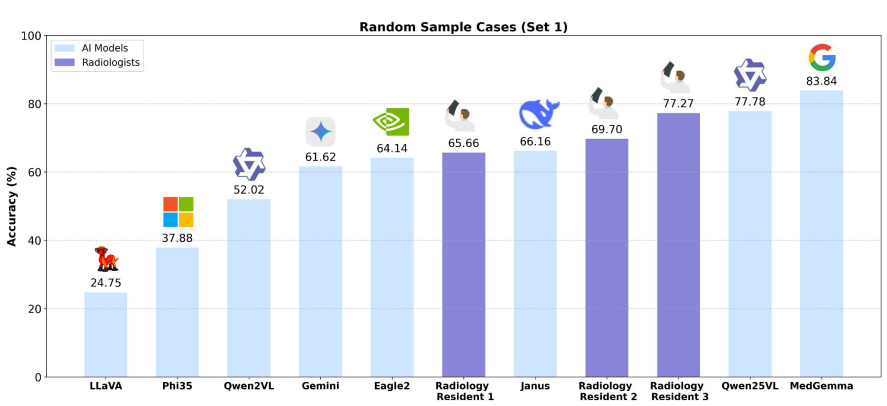
Contributions

1. We created the **largest and most comprehensive medical VQA benchmark** with **694,841 questions** evaluating five distinct cognitive abilities that mirror clinical workflows, using expert-refined prompts validated by radiologists.
2. We developed a rigorous **three-layer pipeline** combining computational processes with expert oversight, ensuring clinical authenticity and reducing template-based artifacts common in existing datasets.
3. We demonstrated the first instance where AI performance consistently exceeds expert human evaluation in chest X-ray interpretation, with MedGemma achieving 83.84% vs. best radiologist at 77.27%.

Results

MedGemma achieves exceptional performance at 83.24% overall accuracy, establishing a new benchmark for multimodal medical image interpretation.

Model	Overall Accuracy ↑	Differential Diagnosis ↑	Geometric Information ↑	Location and Distribution ↑	Negation Assessment ↑	Presence Assessment ↑	Failed Extractions ↓
LLaVA	26.61	21.61	23.46	27.61	24.02	36.33	2.37
Phi35	47.49	62.24	22.15	37.11	79.50	36.44	0.05
Qwen2VL	54.70	52.65	44.94	54.05	62.69	59.15	0.01
Gemini	63.31	62.21	46.89	59.60	85.68	62.17	**0.0**
Eagle2	64.43	68.17	56.98	56.95	**86.32**	53.75	**0.0**
Qwen25VL	65.55	63.61	66.48	63.24	83.27	51.14	**0.0**
Janus-Pro-7B	66.56	56.34	75.42	64.62	75.73	60.70	**0.0**
MedGemma	**83.24**	**76.71**	**80.45**	**83.47**	85.03	**85.21**	**0.0**

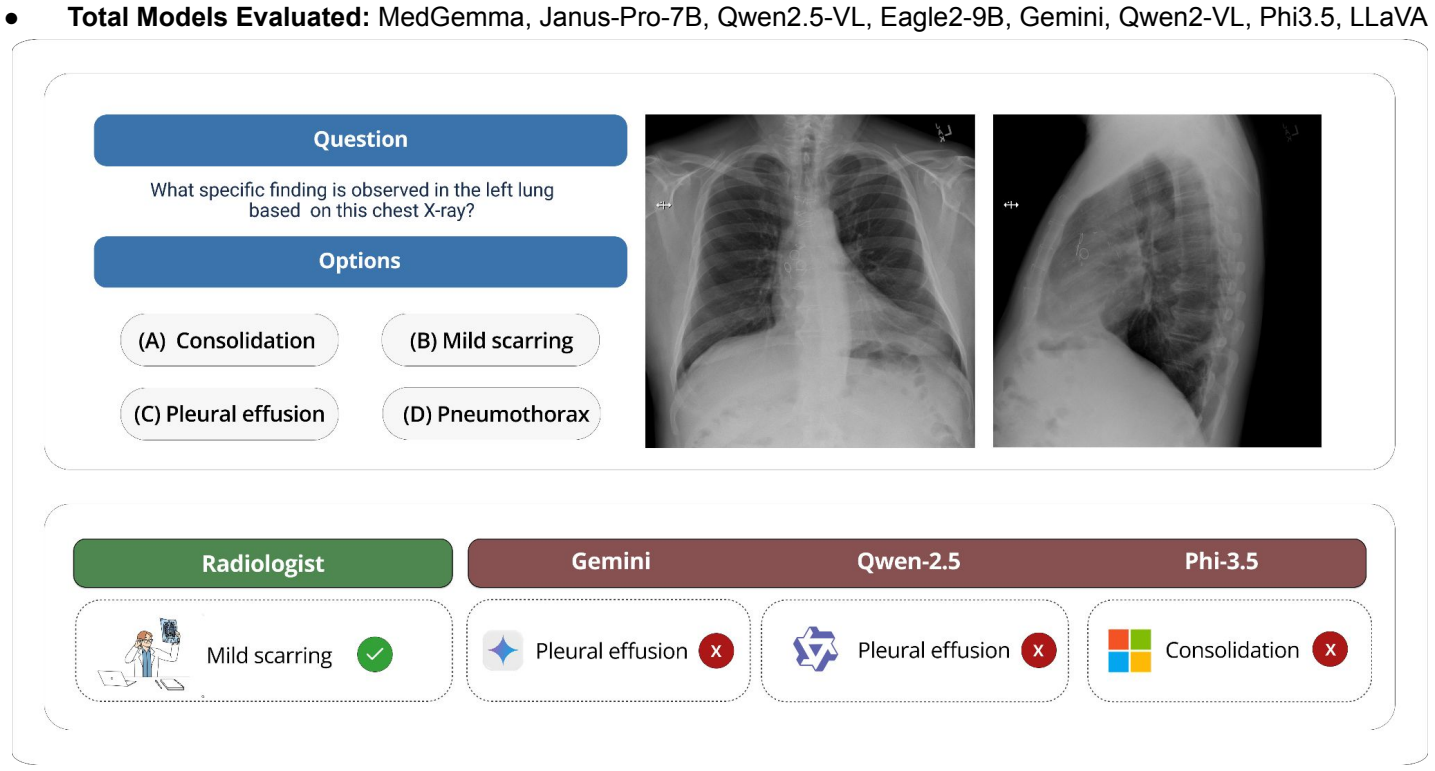


MedGemma leads in **four out of five major categories**, demonstrating superior clinical reasoning across diverse radiological tasks.

Category	Gemini	Eagle2	Janus	LLaVA	Q2VL	Q25VL	Phi35	MedGemma
Clinical Assessment								
Quality of Exams	70.30	62.50	51.64	10.30	40.50	60.92	35.10	**71.23**
Respiratory System								
Lung & P.O	72.24	72.19	68.70	32.98	60.73	64.77	58.14	**80.44**
Lung & P.L	80.00	64.65	58.59	19.34	61.62	78.64	58.82	**87.88**
Lung Volume	65.44	58.22	51.64	39.72	52.11	60.92	28.25	**78.64**
Cardiovascular								
Heart	80.29	81.71	72.01	26.01	60.31	84.83	62.73	**97.03**
Aorta	76.65	41.04	60.14	6.84	72.17	34.79	33.05	**87.86**
Other Great Vessel	**73.33**	60.00	60.00	13.33	60.00	60.00	45.45	**73.33**
Medical Devices								
Tubes and Lines	59.45	58.26	58.87	22.81	48.78	65.04	32.10	**83.86**
Implanted Devices	54.46	52.64	60.79	29.50	52.40	53.48	43.16	**73.14**
Pathologies								
Infectious Disease	71.55	63.24	66.77	56.63	51.02	67.74	42.13	**77.61**
Pulmonary Neoplasm	78.46	66.92	81.95	39.29	63.91	73.68	69.44	**88.72**
Negation	61.05	71.73	58.19	15.00	56.37	61.49	**78.96**	74.76
Musculoskeletal								
Rib	88.90	83.93	89.80	36.92	86.35	79.21	78.56	**91.84**
Spine	78.84	62.30	86.43	64.84	73.73	50.10	57.94	**92.68**
Clavicle	75.93	57.14	75.00	23.21	71.43	33.93	51.02	**92.73**
Joint	51.96	42.31	70.19	47.06	66.35	36.54	42.22	**88.46**
Average	74.00	67.00	67.00	38.00	64.00	66.00	50.00	**83.24**

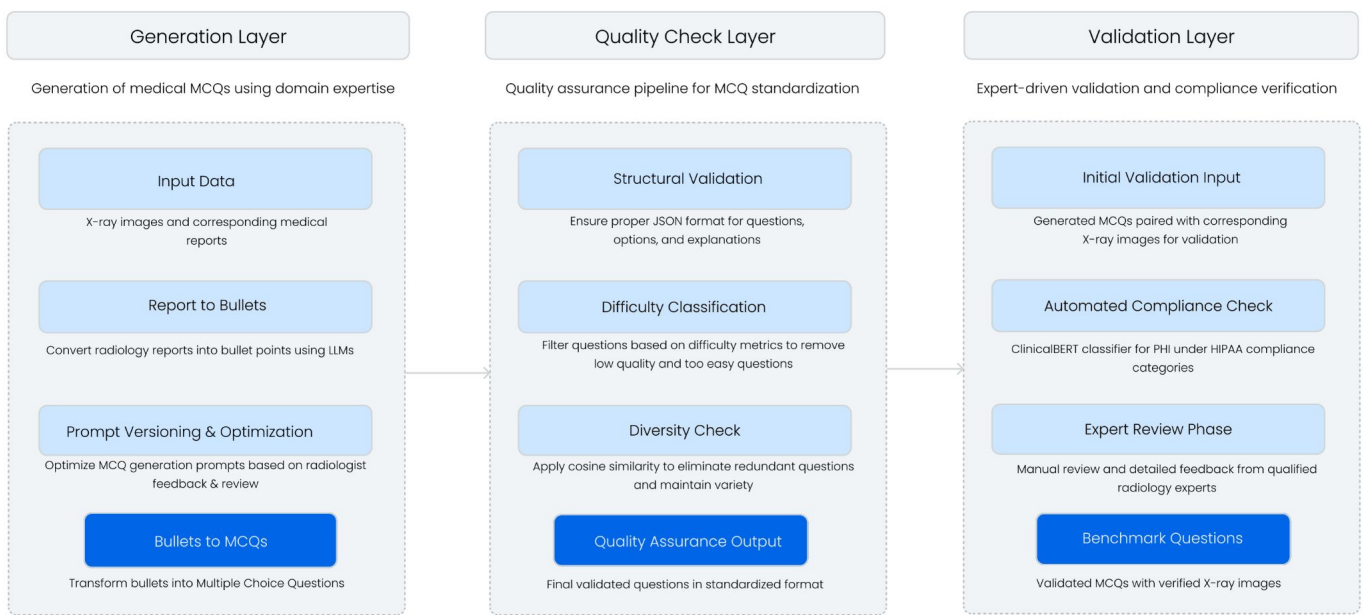
Challenge Design

- **Dataset Scale:** 694,841 multiple-choice questions paired with 160,000 chest X-rays
- **Participants:** 8 state-of-the-art multimodal large language models
- **Task:** Visual question answering across five core radiological reasoning skills
- **Evaluation Framework:** Negation Assessment (36.5%), Presence Assessment (36.1%), Differential Diagnosis (20.9%), Location & Distribution Assessment (6.1%), Geometric Information Analysis (0.4%)
- **Total Models Evaluated:** MedGemma, Janus-Pro-7B, Qwen2.5-VL, Eagle2-9B, Gemini, Qwen2-VL, Phi3.5, LLaVA



Data Pipeline

- **ReXGradient-160K:** 170,000 chest X-ray studies from 4 U.S. health systems
- **Expert-Guided Generation:** Three-layer pipeline with radiologist validation
- **Equipment Diversity:** SIEMENS, FUJI, SAMSUNG, VIDAR, TOSHIBA, GE across all sites



Key Findings

- **Medical Specialization Advantage:** Specialized medical training provides substantial improvements over general-purpose models, with MedGemma outperforming the next best model by 16.68% overall accuracy
- **Task-Specific Capabilities:** Different cognitive skills require explicit design rather than uniform diagnostic reasoning approaches, as MedGemma achieves 85.03% on negation tasks but only 76.71% in differential diagnosis
- **Clinical Authenticity:** Expert-guided dataset creation produces more representative questions than template-based approaches, with radiologist validation reducing errors from 10.8% to under 1% in final questions
- **Human-AI Performance Gap:** AI consistently exceeds expert human evaluation in standardized assessment, marking the first instance of AI surpassing radiologist performance (83.84% vs 77.27%)